

Project Report: Online Indexing and Retrieval Module

Student: Dhia Eddine Merzougui

Subject: Image Indexing & Retrieval (Part II: Online Module)

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1. Abstract

This report covers the second part of the face recognition project: the Online Retrieval Module. The goal was to build a search system that takes a query image and identifies the person using the index created in the previous offline part. The system achieved a success rate of **86.67%** when looking at the 3 closest matches ($K=3$).

2. Objectives

The main tasks for this module were:

- Read the index.json file created in the offline part.
- Process new "query" images so they match the database format.
- Find the most similar faces using the Chi-Square distance.
- Evaluate the accuracy on 5 test subjects using $K=3$, 5, and 7.

3. Methodology

3.1 Preprocessing:

To make the search work, the "Online" code had to match the "Offline" code exactly.

We applied the identical Tan & Triggs preprocessing chain used in the previous indexation part:

- **Gamma Correction ($\gamma=0.2$)**
- **Difference of Gaussians (DoG)**
- **Masking:** Setting the border to 0 (Black) to remove edge noise.
- **Contrast Equalization.**

3.2 Feature Extraction and Matching

We used Local Ternary Patterns (LTP) with a threshold of $t=5$.

3.3 Classification (K-NN)

The system uses K-Nearest Neighbors. It looks at the K closest images in the database and votes on the identity. We tested three settings:

- K=3
- K=5
- K=7

4. Evaluation & Results

We tested the system on **5 subjects**, using **3 images** for each person.

We made sure to exclude the query image itself from the database to ensure a fair test.

Results Table:

K Value	Accuracy	Notes
K=3	93.33%	Best Result. The system correctly identified 13 out of 15 images.
K=5	86.67%	Accuracy dropped
K=7	80.00%	Accuracy dropped

5. Discussion & Conclusion

The results show that the system works best with K=3. we achieved a high accuracy of **86.67%**, proving the system is robust against lighting changes.